Dissolved Oxygen Prediction for Anyang Stream

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Research objective and contribution

In the Smart Water Environment Management course, I developed a dissolved oxygen (DO) prediction workflow for Anyang Stream. I compared missing-data treatments and six predictive models, then tuned a long short-term memory (LSTM) network. The optimized LSTM achieved $R^2 = 0.915$ on the test set and $R^2 = 0.966$ on subsequent November 2024 observations.

Data and missing value analysis

The historical dataset contained 17,965 water-quality records from November 2022 through October 2024. DO was the prediction target, with water temperature, pH, total nitrogen (TN), total phosphorus (TP), and total organic carbon (TOC) as five predictors. Figure 1 shows the observed missing-data patterns across these variables. I compared mean imputation, multivariate imputation by chained equations (MICE), k-nearest-neighbor (KNN) imputation, and an ensemble averaging MICE and KNN estimates. Under simulated missing completely at random (MCAR) rates of 10%, 20%, and 30%, I assessed reconstruction using RMSE, MAE, and R^2. The MICE–KNN ensemble was selected based on these comparisons for the prediction workflow.

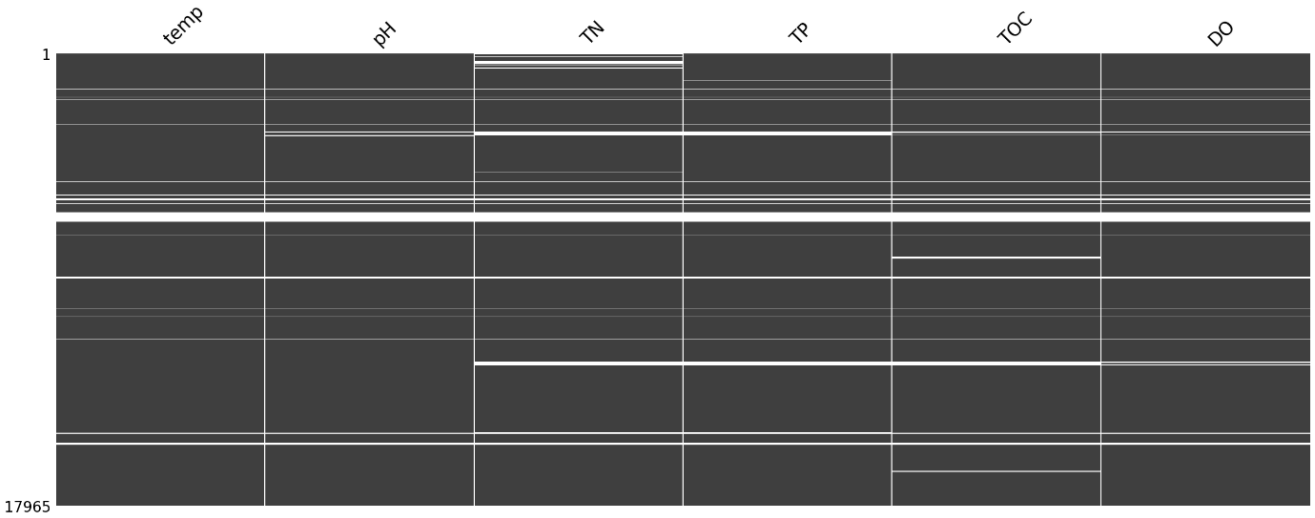


Figure 1. Observed missing-data patterns. Dark regions indicate available records and white gaps indicate missing values. Reproduced from the original course report, Fig. 4, p. 12.

Model comparison and LSTM tuning

I compared multiple linear regression, random forest, LightGBM, LSTM, gated recurrent unit (GRU), and a Transformer model. This comparison examined both conventional machine-learning methods and neural networks for predicting DO from the water-quality data. LSTM was then selected for further hyperparameter tuning.

I tested LSTM unit counts of 50 and 100, batch sizes of 16, 32, 64, 128, and 256, and dropout rates of 0.1, 0.2, and 0.3, using early stopping during training. The selected configuration used 100 units, a batch size of 32, and a dropout rate of 0.1.

Test set evaluation

The tuned LSTM achieved a test-set R^2 of 0.915. Figure 2 compares its predictions with the observed DO values. The original plot uses scaled DO values and time-step indices.

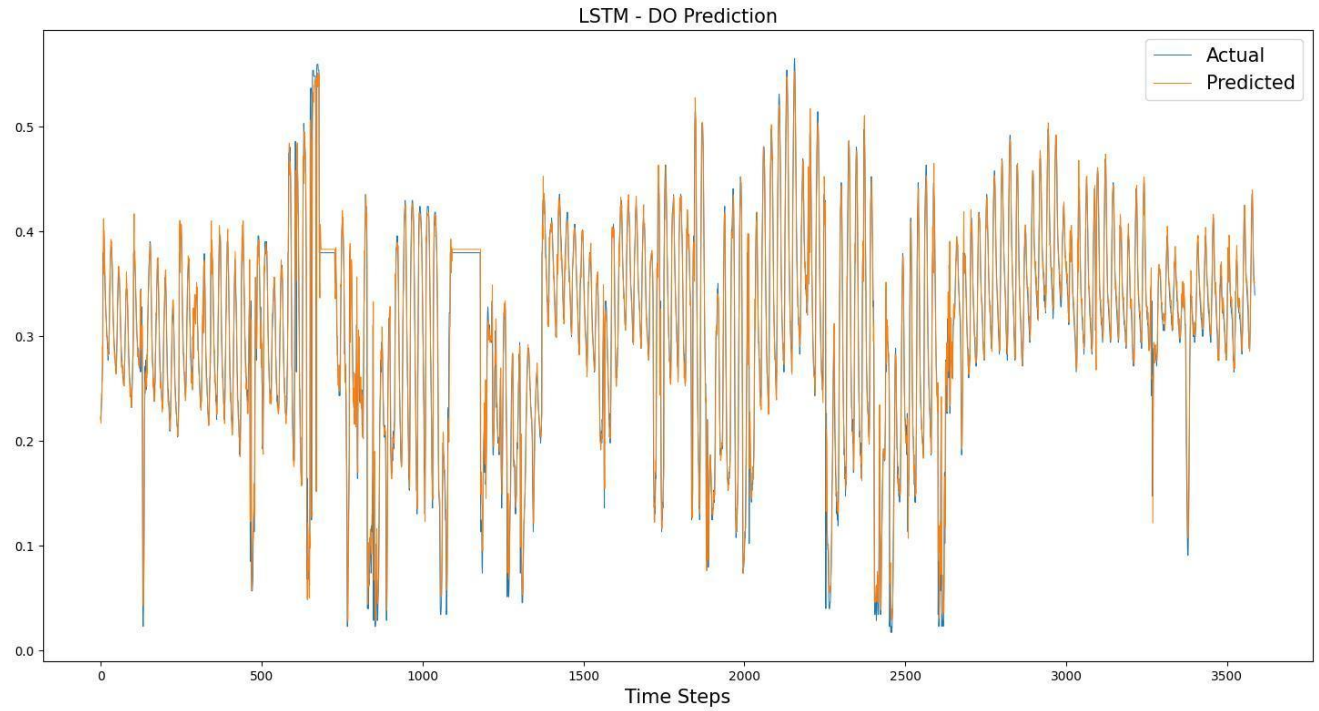


Figure 2. Actual and predicted scaled DO for the tuned LSTM on the test set, $R^2 = 0.915$. Reproduced from the original course report, Fig. 16, p. 24.

The predictions closely followed much of the observed variation, including many sharp changes. This result describes performance on the assessed test set; it does not by itself establish accuracy during low-oxygen events or suitability for real-time monitoring. Evaluation on subsequent observations provided a separate check beyond the historical data period.

Evaluation on subsequent observations

I also evaluated the tuned LSTM on November 2024 observations that were not used for training. It achieved R^2 = 0.966 on this later dataset. Figure 3 shows the corresponding actual and predicted scaled DO values, retaining the original report's time-step axis.

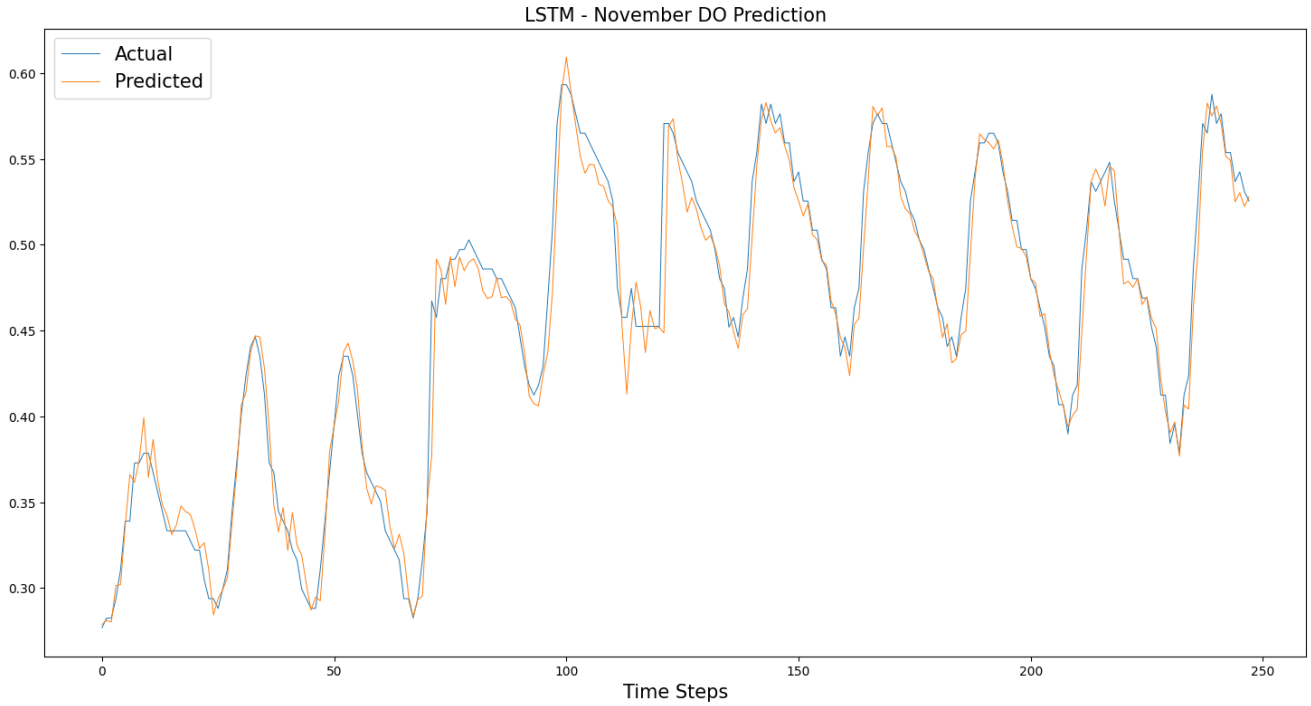


Figure 3. Actual and predicted scaled DO for the tuned LSTM on subsequent November 2024 observations, R^2 = 0.966. Reproduced from the original course report, Fig. 17, p. 25.

Findings and scope

The test-set and November scores describe different datasets, so the higher November R^2 is not a measured improvement on the same cases. The imputation benchmark also simulated MCAR conditions, whereas the observed gaps included structured patterns. These results support the comparative workflow within the evaluated data; further checks across seasons, locations, and realistic sensor outages would be needed to assess broader monitoring use.

Project outcome

The project ranked first among all course projects, was selected as an outstanding project, and was presented at the course showcase. It strengthened my ability to compare methods for incomplete environmental data and to connect model selection with evaluation on both a test set and subsequent observations.